Belief State Geometry in Non-Ergodic Transformers: A Mess3 Mixture Experiment

Abstract

We train a small transformer on a non-ergodic mixture of three Mess3 Hidden Markov Model processes, where each training sequence is generated by one component. This construction requires the model to solve two epistemically distinct tasks: identifying which component is active (meta-belief π) and tracking within-component state (η_k). We pre-register the prediction that residual stream activations encode these in orthogonal subspaces (H2: the hypothesis that π and each η_k are encoded in mutually orthogonal residual stream subspaces). Analysis confirms H2 for the π–η split (removing the π subspace causes $<$0.001 drop in η_k regression R^2), but not for the η_k–$\eta_{k'}$ structure (overlap 0.53–0.69), consistent with superposition. $R^2(\pi)$ rises sharply to near 1.0 by position 2, well before the synchronisation horizon $N^* \approx 9.6$ (Riechers et al., 2025): high linear decodability requires only systematic covariation with component identity across sequences, not full convergence of π to a vertex. Training dynamics reveal η_k representations emerge before π (step 100 vs. 2,000). A dimension sweep shows the model prioritises η_k over π under capacity pressure.

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1 Motivation

Real language corpora are *non-ergodic*: each document belongs to one domain or style and never switches mid-document. A model trained on such data must solve two epistemically distinct tasks: identify *which* type of document it is reading (meta-belief π), and track fine-grained structure within that type (within-component belief η_k). We formalise this with a non-ergodic mixture of $K = 3$ Mess3 HMMs, a clean, analytically tractable proxy where ground-truth belief states are computable exactly.

The experimental design is grounded in three prior results:

- Shai et al. [1] show that a transformer trained on a single ergodic Mess3 process encodes belief states as a fractal in the residual stream.
- Riechers et al. [2] prove that non-ergodic pretraining necessarily produces ICL, governed by a synchronisation horizon N^*.
- Shai et al. [3] show that parallel factors are encoded in orthogonal subspaces, a result for simultaneous streams, not temporal mixtures.

This experiment combines the non-ergodic setup of Riechers et al. with the geometry analysis of Shai et al. to ask: *non-ergodic mixture → factored geometry?* It also provides a geometric complement to Riechers et al.: their synchronisation horizon N^* predicts *when* the model commits to one component; we ask *where* in the residual stream that commitment is encoded.

2 Setup

2.1 Data

Mess3 HMM. A Mess3 process is a 3-state, 3-token edge-emitting Hidden Markov Model parameterised by emission strength $\alpha \in (0.5, 1)$ and mixing rate $x \in (0, 1/3)$. The *token-labeled transition matrix* $T^{(a)} \in \mathbb{R}^{3\times 3}$ encodes the joint probability of emitting token a and transitioning to the next state:

$$T^{(a)}_{ij} = P(\text{token} = a \mid s_t = i) \cdot P(s_{t+1} = j \mid s_t = i),$$

where $P(\text{token} = a \mid s = i) = \alpha$ if $a = i$, else $(1 - \alpha)/2$, and $P(s' = j \mid s = i) = 1 - 2x$ if $i = j$, else x. Sequences are generated by starting in a state drawn from the (uniform) stationary distribution, then at each step emitting a token according to the current state and transitioning to the next state jointly.

Components. We use three Mess3 HMMs with the following parameters (Table 1):

Component	α	x	Character
A	0.95	0.02	Sharp fractal, slow mixing
B	0.80	0.08	Medium fractal, moderate mixing
C	0.55	0.25	Sparse attractor, fastest mixing

Table 1: Mess3 component parameters. α is emission strength; x is the off-diagonal transition probability.

Parameters were chosen so that: (a) components are distinguishable within 16 tokens ($N^* \approx 9.6 < 16$), and (b) each component produces a distinct fractal geometry in belief space. Figure 1 shows the Mixed State Presentation (MSP) attractor for each component: the fractal attractor of the belief state iterated function system (IFS).

Sequence generation. To form the training corpus, a component $k \sim \text{Uniform}(\{A, B, C\})$ is sampled for each sequence. A token sequence $x_{1:L}$ of length $L = 16$ (the context window) is then generated from component k alone, starting from the stationary distribution. The context window was set to $L = 16$ to satisfy $N^* < L$: with $N^* \approx 9.6$, the model accumulates enough context to synchronise to the correct component within a single sequence. The corpus is *non-ergodic* because sequences never switch components; each sequence is a pure draw from one Mess3 process. Figure 2

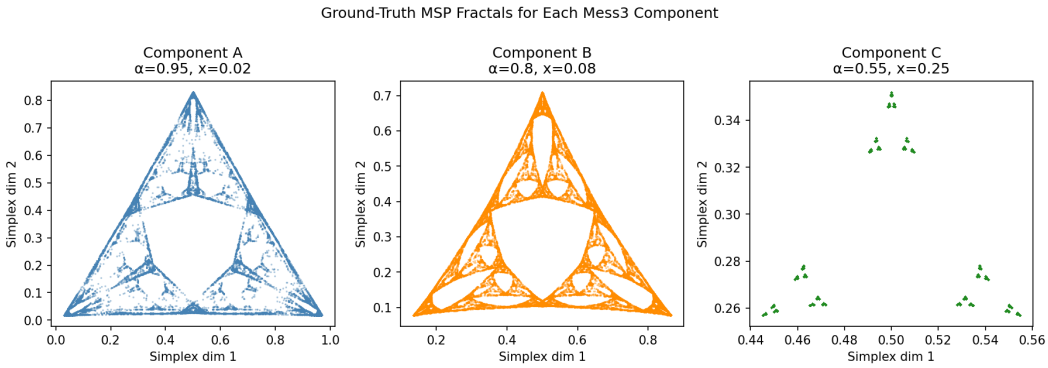


Figure 1: Ground-truth MSP fractals for each Mess3 component, simulated by iterating the IFS $f_a(\eta) = \eta T^{(a)}/(\eta T^{(a)}\mathbf{1})$ for 80,000 steps. Component A (blue) produces a rich self-similar fractal; component C (green) has only ∼9 reachable belief states due to weak emission ($\alpha = 0.55$). These three geometries define the expected residual stream structure under H2.

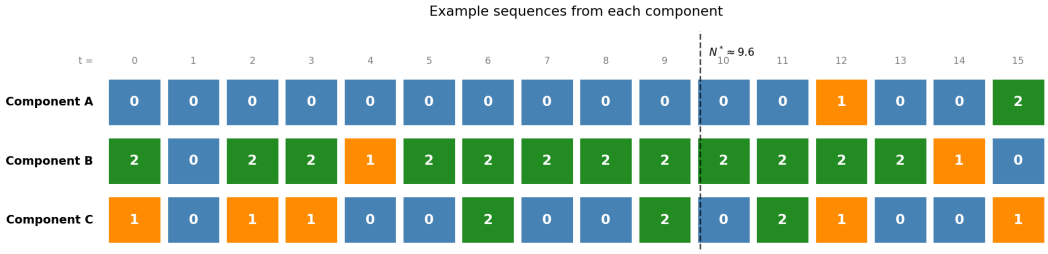


Figure 2: One example sequence (length 16) from each component. Cell color encodes token value (blue = 0, orange = 1, green = 2). Component A ($\alpha = 0.95$) is nearly always token 0; component C ($\alpha = 0.55$) mixes all three tokens freely. The dashed line marks the synchronisation horizon $N^* \approx 9.6$: before this point the identity of the component is ambiguous; after it a Bayesian observer has committed. The contrast in token diversity across rows reflects the components' different emission strengths and makes them distinguishable within N^* tokens.

shows one example sequence per component; the visual contrast in token diversity across rows reflects the very different α and x values.

Ground-truth belief states. At each position t, the Bayesian-optimal agent maintains a joint belief $(\pi(t), \eta_A(t), \eta_B(t), \eta_C(t))$, where $\pi(t) \in \Delta^2$ is the meta-belief over components and $\eta_k(t) \in \Delta^2$ is the within-component belief for component k. Both are initialised to the uniform stationary distribution (which equals uniform for Mess3 by symmetry) and updated after each observed token using the two-step rule (Section 3.3). These analytically computed belief states are stored alongside each sequence and used as regression targets in the geometry analysis. 50,000 training sequences and 5,000 validation sequences were generated.

Figure 3 illustrates the full joint 9-state belief geometry: the meta-belief π lives on the central simplex, while each vertex hosts the MSP fractal of the corresponding component. The three fractals are disconnected, reflecting the non-ergodic block structure.

2.2 Architecture

We use a small GPT-2-style transformer implemented via TransformerLens [4]: 2 layers, $d_{\text{model}} = 64$, 4 heads, $d_{\text{head}} = 16$, $d_{\text{mlp}} = 256$, context window 16, vocabulary $\{0, 1, 2\}$, ≈101k parameters.

2.3 Training

Cross-entropy next-token prediction, AdamW ($\text{lr} = 10^{-3}$, cosine decay to 10^{-5}, 1,000 warmup steps), 50,000 steps total. Batch size 64 (3,200,000 total sequence exposures over 50,000 steps).

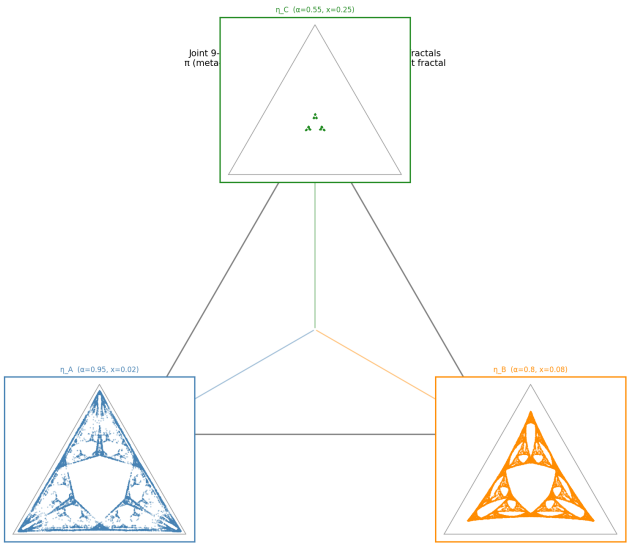


Figure 3: Joint 9-state belief geometry. The centre shows the meta-belief π simplex (which component?); each vertex hosts the MSP fractal attractor of that component (where within?). The three fractals are disconnected (no probability mass flows between components), reflecting the non-ergodic block structure of the corpus.

Sequences are generated on-the-fly each step to prevent overfitting. Checkpoints saved at steps 100, 500, 1,000, 2,000, then every 5,000 steps; at each checkpoint we run regression at the final layer and final position to track how R^2 evolves over training.

3 Hypotheses and Predictions

3.1 Possible Hypotheses

H1 (Joint): The transformer represents η_{joint} as one undifferentiated 8D blob. High R^2 for all targets, but projecting out the π subspace substantially degrades η_k regression.

H2 (Factored): Four orthogonal subspaces: one 2D π subspace and three 2D η_k subspaces. Projecting out π leaves η_k $R^2 \approx$ unchanged.

H3 (Superposition): π and η_k recoverable approximately but packed into fewer than 8 effective dimensions via near-orthogonal superposition.

3.2 Intuitive Prediction

Our geometric intuition is that the residual stream should mirror the ground-truth joint belief geometry: the *nested fractal* shown in Figure 3, with π occupying a central simplex and each vertex of that simplex hosting the MSP fractal attractor of the corresponding component.

We favour H2 (Factored) by analogy to Shai et al. [3], where the setting is a *parallel* joint process: at each timestep independent subprocesses each emit a token and the observation is their concatenation. Our setup is *sequential* (each sequence comes from one subprocess), but the underlying math is the same: block-diagonal transition matrices, conditionally independent factors, and a belief space that decomposes into orthogonal subspaces. If transformers learn factored representations in the parallel case, we expect them to do so here too. We therefore pre-register H2: π and each η_k are encoded in mutually orthogonal residual stream subspaces, with CEV dimensionality ≈ 8 (four 2D subspaces: one for π, one each for η_A, η_B, η_C).

3.3 Mathematical Derivation

The joint belief state. The joint hidden state space is $\mathcal{S}_{\text{joint}} = \mathcal{S}_A \sqcup \mathcal{S}_B \sqcup \mathcal{S}_C$ (9 states). The joint token-labeled transition matrices are block-diagonal:

$$T^{(x)}_{\text{joint}} = \text{diag}\Big(T^{(x)}_A,\, T^{(x)}_B,\, T^{(x)}_C\Big)\,.$$

The zeros off the block diagonal encode non-ergodicity exactly: no probability mass can flow between components. This block structure means the joint belief state $\eta_{\text{joint}} \in \Delta^8$ (the 9-state probability simplex, affine dimension 8) decomposes uniquely as

$$\eta_{\text{joint}}[k, s] = \pi_k \cdot \eta_k[s],$$

where $\pi \in \Delta^2$ is the *meta-belief* and $\eta_k \in \Delta^2$ is the *within-component belief*.

The two-step update. Upon observing token x:

$$L_k(x) = \eta_k \cdot T^{(x)}_k \cdot \mathbf{1} \quad \text{(per-component likelihood)}, \tag{1}$$

$$\pi'_k = \frac{\pi_k \cdot L_k(x)}{\sum_j \pi_j L_j(x)} \quad \text{(meta-belief Bayes update)}, \tag{2}$$

$$\eta'_k = \frac{\eta_k \cdot T^{(x)}_k}{L_k(x)} \quad \text{(within-component Bayes update)}. \tag{3}$$

Crucially, equation (3) applies the standard single-Mess3 update *independently* to each component. The components couple only through the normalisation constant in (2).

Synchronisation horizon. The *synchronisation horizon* N^* is the characteristic number of tokens required for the meta-belief $\pi(t)$ to collapse near the vertex of the true component, given by

$$N^* = \frac{1}{\min_{k \neq k'} D_{\text{KL}}(P_k \| P_{k'})},$$

where D_{KL} is estimated empirically via per-token log-likelihood ratios over long sequences. With the chosen parameters, $N^* \approx 9.6 < 16$, so synchronisation completes within every training sequence.

4 Methods

PCA and CEV dimensionality. To measure how many dimensions the transformer uses to represent the belief state, we run PCA on the matrix of all residual stream vectors per layer, flattening sequences and positions into rows to obtain an $(N \cdot L) \times d_{\text{model}}$ matrix. The principal components are ordered by variance explained. The **CEV (cumulative explained variance) dimensionality** at threshold p is the smallest d such that the top-d components together explain $\geq p\%$ of total variance: the minimum number of directions needed to capture $p\%$ of the structure in the activations.

Regression-based subspace identification [1]. We fit a linear map $W \in \mathbb{R}^{3 \times d_{\text{model}}}$ from residual stream activations $\mathbf{a} \in \mathbb{R}^{d_{\text{model}}}$ (normalised to zero mean and unit variance per dimension) to ground-truth belief targets $\mathbf{b} \in \mathbb{R}^3$ (either π or η_k) via Ridge regression ($\lambda = 10^{-4}$), and measure linear decodability by R^2 on a held-out 20% test split (fit on 80%). To identify *where* in the residual stream the belief is encoded, we decompose $W = U\Sigma V^\top$, where the columns of V are orthonormal directions in activation space ordered by their contribution to the prediction. Since belief states live on a 2-simplex (affine dimension 2), we take the top-2 columns of V as the belief-encoding subspace. This basis is used in all subsequent overlap tests.

Subspace overlap metric [3]. Given two subspaces with orthonormal bases $Q_A \in \mathbb{R}^{k_A \times d}$ and $Q_B \in \mathbb{R}^{k_B \times d}$ (orthonormal rows), the overlap is

$$\text{overlap}(Q_A, Q_B) = \frac{\|Q_A Q_B^\top\|_F}{\sqrt{k_A \cdot k_B}}.$$

The (i, j) entry of $Q_A Q_B^\top$ is the cosine $\cos\theta_{ij}$ between the i-th basis vector of Q_A and the j-th basis vector of Q_B; the Frobenius norm is the square root of the sum of their squares, and dividing by $\sqrt{k_A k_B}$ normalises to $[0, 1]$ (0 = orthogonal subspaces, 1 = identical subspaces).

Fractal visualisation. We visualise whether the residual stream encodes the full fractal geometry of each component's belief attractor, not merely a summary statistic. For each component k, we take residual stream activations from component-k sequences at positions $t > N^*$ (after the meta-belief π has collapsed, so activations reflect pure component-k belief states), apply the η_k regression map to obtain predicted $\hat{\eta}_k \in \mathbb{R}^3$, and plot the resulting point cloud in barycentric simplex coordinates. We compare this cloud against the ground-truth MSP attractor, generated by iterating the IFS $f_a(\eta) = \eta T^{(a)}/(\eta T^{(a)}\mathbf{1})$ for 80,000 steps.

5 Results

5.1 PCA Dimensionality

Layer	CEV (95%)	CEV (98%)	CEV (100%)	Predicted
0	6	9	64	8
1	5	7	64	8

Table 2: PCA dimensionality per layer. CEV(p%) = fewest components explaining $\geq p$% of variance. The first 5–6 dims capture 95%, 7–9 capture 98%, and all 64 carry non-zero variance, revealing a low-dimensional dominant subspace atop a long noise tail.

Table 2 reveals a striking compression: the transformer uses fewer dimensions than the theoretical belief space requires. Theory predicts 8 dimensions (four 2D subspaces), yet 98% of variance is captured in just 7 dimensions at the final layer, and 95% in only 5.

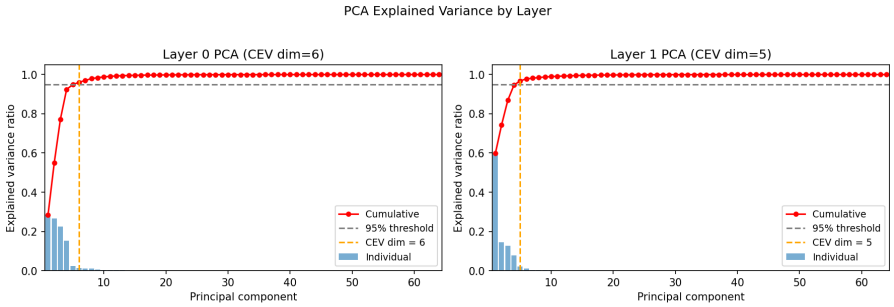


Figure 4: Cumulative explained variance (CEV) by context position, for Layer 0 (dashed) and Layer 1 (solid). The predicted 8 dimensions under H2 are marked by the dotted vertical line. Layer 1 explains 95% of variance in just 5 components, more compressed than the theoretical minimum, consistent with fractal sparsity and superposition.

Figure 4 plots this. This suggests the transformer is *more efficient* than the theoretical minimum representation. We offer three possible explanations, though we cannot confirm which dominates:

1. **Component C's near-degenerate attractor.** With $\alpha = 0.55$, component C has only $\sim$9 reachable belief states, collapsing η_C from a 2D subspace toward a near-discrete structure that requires far less than 2 dimensions.
2. **Fractal attractors don't fill their 2D subspaces uniformly.** Each MSP attractor is a fractal with dimension strictly less than 2. Belief states cluster along lower-dimensional structures within each 2D subspace, so PCA sees less than 2 effective dimensions per factor.
3. **Superposition enabled by attractor sparsity.** Because the attractors are sparse, the transformer can pack multiple belief directions into shared dimensions without interference, achieving near-perfect $R^2 > 0.95$ with fewer dimensions than a uniform distribution would require.

At 100% CEV both layers require all 64 dimensions, confirming a sharp transition between the dominant low-dimensional belief subspace and a long tail of noise dimensions.

5.2 Linear Decodability

Quantity	R^2
π (meta-belief)	0.957
η_A (component A)	0.950
η_B (component B)	0.963
η_C (component C)	0.984

Table 3: Ridge regression R^2 at the final layer, final context position (position 15).

All belief targets are linearly decodable with $R^2 > 0.95$ [7], confirming that the residual stream encodes the full joint belief state.

Figure 5 shows R^2 as a function of context position. $R^2(\eta_k)$ is near 1.0 from position 1, confirming that within-component belief is encoded reliably from the very first token. $R^2(\pi)$ starts at 0 (no context at position 1) and rises sharply to near 1.0 by position 2, remaining stable thereafter. The aggregate $R^2(\pi) = 0.956$ in Table 3 reflects this stability averaged over all positions.

This early rise indicates the model has learned the emission statistics of each component during training. The first token is already highly diagnostic: token 0 is emitted with probability $\alpha_A = 0.95$ under component A but only ≈ 0.3 under component C, so the Bayesian posterior over components shifts substantially after a single observation. The model encodes this partial posterior accurately and immediately.

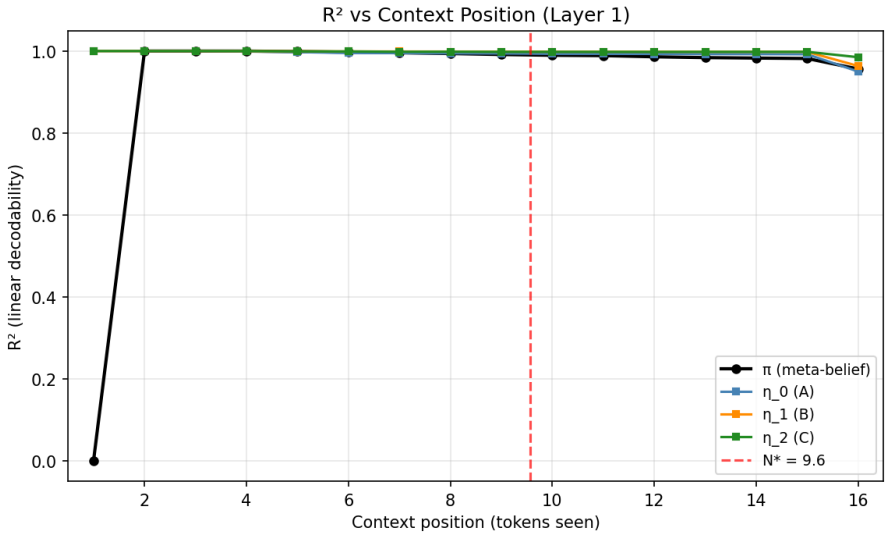


Figure 5: R^2 vs. context position for Layer 1. $R^2(\eta_k)$ (coloured) is near 1.0 from position 1. $R^2(\pi)$ (black) starts at 0 and rises sharply to near 1.0 by position 2.

5.3 Subspace Orthogonality

π is nearly orthogonal to all η_k subspaces (overlap 0.17–0.19). The moderate η_k–$\eta_{k'}$ overlap (0.53–0.69) reflects the shared 3-token alphabet: all components respond to the same tokens, inducing partially shared within-component directions, consistent with superposition [5].

As a direct functional test, removing the π subspace from activations and re-running η_k regression yields R^2 drops of less than 0.001 in all cases, and removing any η_k subspace leaves π regression unchanged, confirming that the subspaces are functionally independent, not merely geometrically near-orthogonal.

	π	η_A	η_B	η_C
π	1.000	0.193	0.171	0.176
η_A	0.193	1.000	0.691	0.534
η_B	0.171	0.691	1.000	0.590
η_C	0.176	0.534	0.590	1.000

Table 4: Subspace overlap matrix ($0 =$ orthogonal, $1 =$ identical). π is nearly orthogonal to all η_k subspaces (0.17–0.19). The η_k subspaces overlap each other moderately, reflecting the shared token alphabet.

5.4 Fractal Geometry

Figure 6 shows, for each component, the ground-truth belief state trajectories (left) and the predicted η_k obtained by applying the regression weights to residual stream activations (right), both plotted in barycentric simplex coordinates.

The regression output faithfully reproduces the fractal structure of each component's MSP attractor, confirming that the model encodes the full geometric structure of the belief distribution, not merely its mean. Component A exhibits a rich self-similar fractal; component B a moderately nested fractal; component C a sparse set of $\sim$9 discrete clusters.

5.5 Training Dynamics

Figure 7 shows that η_k subspaces emerge *before* the π subspace.

The ordering (within-component beliefs first, meta-belief later) suggests that the transformer first learns to track the local token statistics of each component individually (which requires only learning the Mess3 update within each component's context), and only subsequently learns to *distinguish* which component is active (which requires comparing cross-component likelihoods). This ordering makes sense: per-component prediction reduces loss immediately, while improving π only helps once the model has already learned the within-component structure to compare.

6 Additional Analysis: Dimension Sweep

Motivation. The main experiment uses $d_{\text{model}} = 64$, giving the model roughly $8\times$ more residual stream dimensions than the theoretical minimum for a full factored representation ($K + 1$ subspaces of 2D each $= 8$ dimensions for $K = 3$). This raises the question: is the inductive bias toward factored representations an artifact of surplus capacity, or does it persist under pressure?

Separately, we ask which belief state the model prioritises when it *cannot* represent both. The factored representation result of Shai et al. [3] says π and η_k are equally important in the sense of conditional independence, but they are not equally important for the *next-token prediction task*. Knowing π alone does not determine the next-token distribution; one also needs η_k for the active component. Conversely, η_k (plus implicit knowledge of which component is active from context) is sufficient. After position N^*, the context itself encodes component identity, making π in the residual stream redundant; η_k must be maintained throughout.

We therefore expect that under capacity pressure the model encodes η_k first.

Setup. We train four additional models with $d_{\text{model}} \in \{2, 4, 6, 8\}$, keeping all other hyperparameters fixed (2 layers, $n_{\text{heads}} \times d_{\text{head}} = d_{\text{model}}$ throughout, see Table 5). Each model is trained for 20,000 steps on the pre-saved 50,000 training sequences via a DataLoader (shuffled, cycled), which avoids the per-step HMM generation overhead that dominates for tiny models. Note that the main $d_{\text{model}} = 64$ model was trained on sequences generated on-the-fly (50,000 steps); the sweep models use the pre-saved 50,000 training sequences via a `DataLoader` (20,000 steps). Val losses are comparable across all models, confirming convergence. Note $d = 8$ is the theoretical minimum capacity for a full factored representation.

Results.

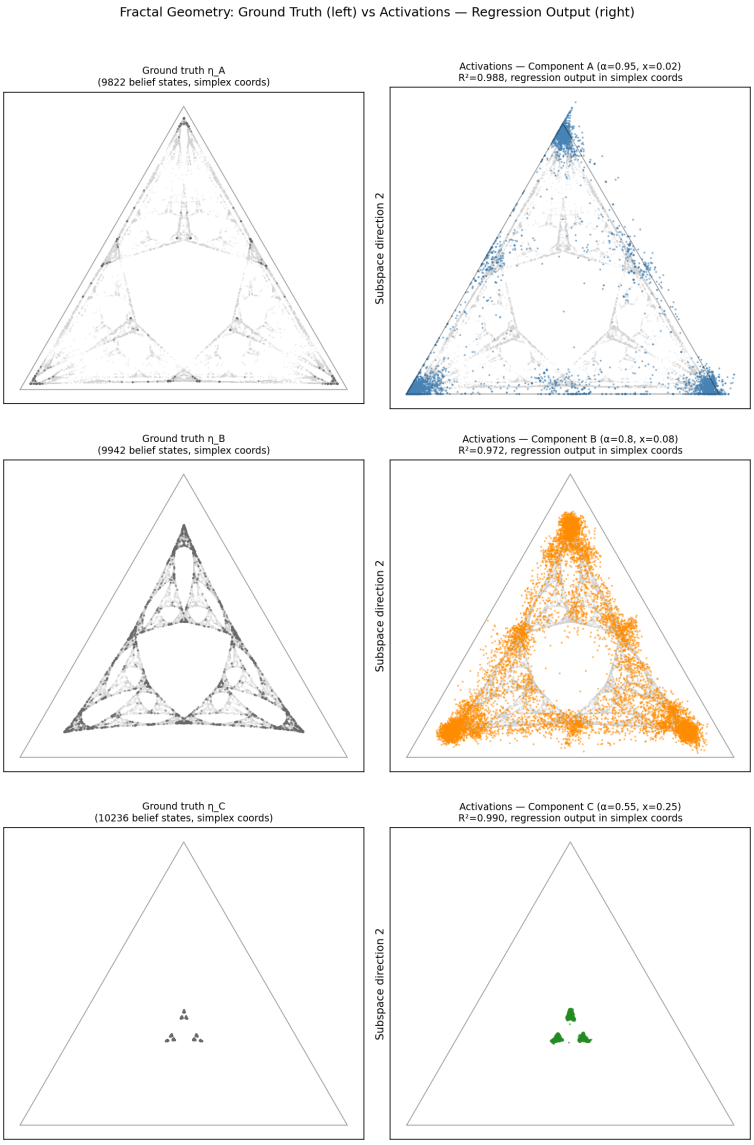


Figure 6: Fractal comparison for each component: ground-truth η_k trajectories (left) vs. regression-predicted η_k from residual stream activations (right), plotted in barycentric simplex coordinates. The regression output reproduces the MSP fractal geometry.

Figure 8 and Table 6 tell a clear story. η_k encoding saturates early: $R^2(\eta_k) \geq 0.80$ already at $d = 4$, and improves only marginally thereafter. By contrast, $R^2(\pi)$ starts near zero and grows slowly, reaching only 0.69 at $d = 8$ (the theoretical minimum for a full factored representation), and 0.957 only at $d = 64$.

The scatter plots in Figure 9 confirm this geometrically. At $d = 4$, PC projections coloured by component identity (left) show a single indistinguishable blob (the model has essentially zero π encoding), while the same projections coloured by dominant within-component state (right) form clean diagonal bands. At $d = 6$, the first signs of component separation appear, but the η_k dip noted above is also visible: within-component structure is slightly degraded relative to $d = 4$, before recovering at $d = 8$.

We note that at $d = 6$, $R^2(\eta_k)$ *drops* slightly relative to $d = 4$ (e.g. η_A: $0.795 \rightarrow 0.771$), recovering at $d = 8$. This suggests that when the model first gains enough capacity to start encoding π ($d = 6$), it does so partly at the expense of within-component encoding, before recovering both at $d = 8$.

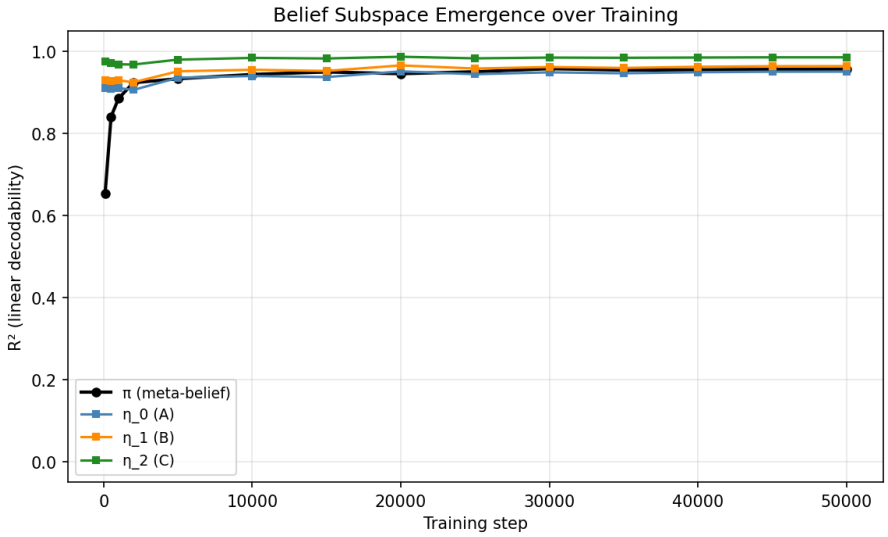


Figure 7: R^2 of π and η_k regressions over training steps (final layer, final position). η_k representations emerge early (step 100, $R^2 > 0.9$); π representation emerges later (step 2,000). The ordering reflects loss-gradient efficiency: per-component prediction reduces loss immediately, while distinguishing components requires cross-component comparison [8].

d_{model}	n_{heads}	d_{head}	d_{mlp}	Params	Val loss
2	1	2	8	199	1.087
4	2	2	16	587	0.880
6	2	3	24	1,167	0.862
8	4	2	32	1,939	0.863
64	4	16	256	101,507	0.862

Table 5: Model configurations for the dimension sweep. Val loss at $d = 6$ already matches $d = 64$; the task is saturated at 6 dimensions.

Interpretation. The result confirms the prediction: η_k is prioritised over π under capacity pressure. The explanation is information-theoretic. The residual stream should encode what the context does not already make obvious. π collapses after $N^* \approx 9.6$ tokens; the context sequence implicitly carries the component identity after that point. η_k, by contrast, requires ongoing Bayesian integration and never becomes obvious from the raw tokens; it must be maintained explicitly in the residual stream throughout the sequence. Furthermore, η_k directly determines the next-token distribution ($P(\text{next} = a \mid \eta_k) = \eta_k \cdot T_k^{(a)} \cdot \mathbf{1}$ for each token a, after synchronisation), while π is a routing variable whose gradient signal vanishes once synchronisation is complete. At small d, the loss gradient therefore drives the model to encode η_k first, leaving π as a low-priority secondary representation.

Together, the four analyses above establish a consistent picture of the residual stream geometry, which we now interpret against the competing hypotheses.

7 Discussion

7.1 Which Hypothesis Best Fits?

The picture is mixed: H2 holds cleanly for the π–η split but not for the internal structure of the η_k subspaces.

H2 confirmed for π vs. η_k. The projection tests are decisive: removing the π subspace from activations causes R^2 drops of less than 0.001 for all η_k targets, and removing any η_k subspace leaves π regression unchanged. The π–η_k subspace overlaps (0.17–0.19) confirm near-orthogonality. The

d	$R^2(\pi)$	$R^2(\eta_A)$	$R^2(\eta_B)$	$R^2(\eta_C)$	CEV dim
2	-0.003	0.384	0.387	0.449	2
4	0.148	0.795	0.820	0.851	3
6	0.649	0.771	0.799	0.745	4
8	0.686	0.830	0.853	0.865	4
64	0.957	0.950	0.963	0.984	5

Table 6: Held-out R^2 for all belief state targets and PCA CEV dimensionality, at the final layer and final context position.

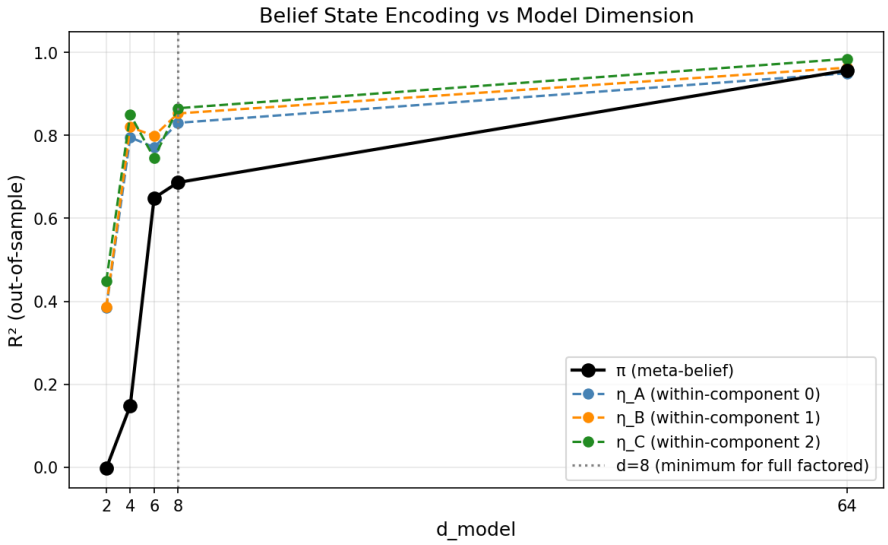


Figure 8: Held-out R^2 vs. d_{model} for all belief state targets. η_k (dashed) saturates by $d = 4$; π (solid black) grows slowly and requires $d \gg 8$ to approach the quality achieved for η_k. The dotted line marks $d = 8$, the theoretical minimum for a full factored representation.

transformer has cleanly separated meta-belief from within-component belief into distinct residual stream directions.

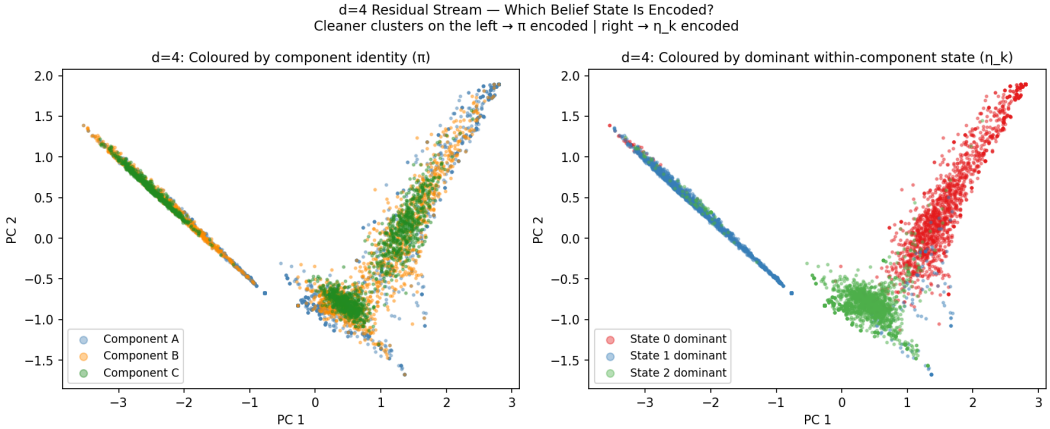
H2 not confirmed for η_k vs. $\eta_{k'}$. H2 predicts four mutually orthogonal subspaces, but the three η_k subspaces overlap substantially with each other (0.53–0.69). This is more consistent with H3 (Superposition): the model does not dedicate a separate 2D subspace to each component's fractal, but instead packs the three within-component beliefs into a shared, partially overlapping region of residual stream space. The vary-one analysis shows that component-specific fits are more distinct (A–C overlap 0.208) than the global fits suggest, so the components are not fully merged, but they are far from orthogonal.

The pre-registered prediction (CEV dimensionality ≈ 8) was not confirmed: both layers use only 5–6 dimensions to explain 95% of variance, consistent with the compressed η_k representation described above.

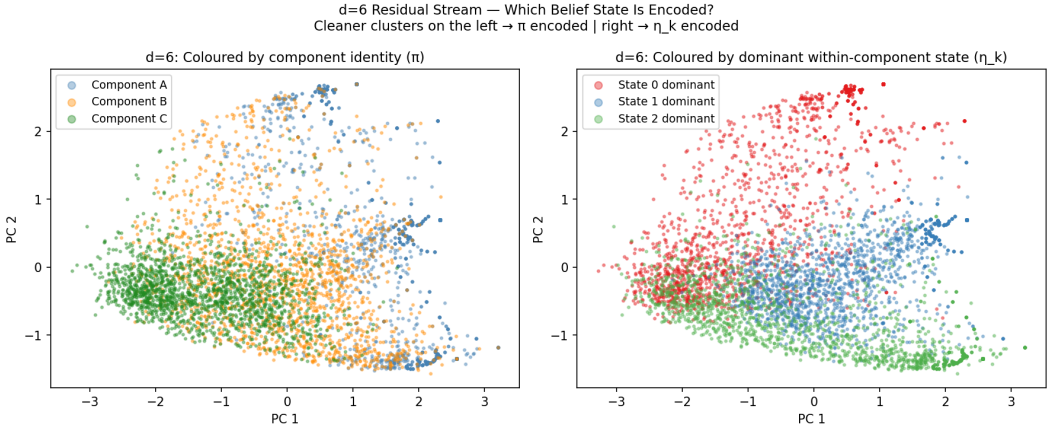
A notable surprise in the training dynamics: within-component beliefs (η_k) emerge early (step 100, $R^2 > 0.9$), while the meta-belief (π) lags behind (step 2,000). This suggests the model first learns to track per-component token statistics (which reduces loss immediately) and only subsequently learns to distinguish *which* component is active (which requires cross-component comparison).

7.2 Implications

Non-ergodic pretraining induces partial factoring. The transformer cleanly separates "what component am I in?" (π) from "where within this component am I?" (η) without explicit supervision. However, the three within-component beliefs are not further separated from each other; they share



(a) $d = 4$: no component separation (left); clear within-component bands (right).



(b) $d = 6$: first sign of π encoding, with slight η_k dip (left); within-component structure preserved (right).

Figure 9: Residual stream PCA projections coloured by component identity (left) and dominant within-component state (right).

residual stream directions, suggesting that factoring happens at the coarsest level of the hierarchy (meta vs. within) but not below it.

Two-level belief structure maps onto two-phase training dynamics. Within-component geometry is learned first and meta-belief routing emerges second; the ordering reflects loss reduction efficiency, not conceptual priority.

In-context learning has geometric structure. ICL in real language models may involve a π-like routing circuit that activates domain-specific η-like representations. The factored geometry found here suggests these circuits can be linearly separated in residual stream space.

7.3 Limitations

The model is small ($d = 64$) with a short context window ($L = 16$), and whether these results generalise to larger architectures remains an open question. The symmetric 3-state/3-token structure of Mess3 makes components easier to distinguish than typical natural language domains, which may make the factored geometry easier to learn than in practice. Our ground-truth belief computation relies on exact analytical updates that are not available for real corpora, so the regression analysis cannot be applied directly to language models without synthetic proxy tasks.

8 Conclusion

A transformer trained on a non-ergodic mixture of Mess3 processes learns a hierarchically factored residual stream geometry: meta-belief π is encoded in a subspace nearly orthogonal to all within-component beliefs η_k, while the η_k subspaces are partially entangled due to their shared geometric structure on the 2-simplex. $R^2(\pi)$ rises sharply by position 2, earlier than the synchronisation horizon $N^* \approx 9.6$ predicted by Riechers et al. [2]: linear decodability requires only systematic covariation with component identity, which the first highly-diagnostic token already provides, while full synchronisation (commitment to one vertex) still requires $\approx N^*$ tokens. Training dynamics show η_k representations emerge an order of magnitude earlier than π (step 100 vs. 2,000), reflecting loss-gradient priority. A dimension sweep confirms this hierarchy persists under capacity pressure: the model sacrifices π encoding before η_k encoding. Future work should test whether these geometric signatures generalise to larger models and richer token alphabets.

References

[1] A. Shai, S. Doshi, F. Langford, and C. J. Hillar. Transformers represent belief state geometry in their residual stream. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[2] P. Riechers, T. McGrath, and A. Tegmark. Next-token pretraining implies in-context learning. arXiv:2505.18373, 2025.

[3] A. Shai, S. Doshi, F. Langford, and C. J. Hillar. Transformers learn factored representations. In *International Conference on Machine Learning (ICML)*, 2025.

[4] N. Nanda and T. Lieberum. TransformerLens: A library for mechanistic interpretability of GPT-2-style language models. https://github.com/neelnanda-io/TransformerLens, 2022.

[5] N. Elhage et al. Toy models of superposition. *Transformer Circuits Thread*, 2022. https://transformer-circuits.pub/2022/toy_model/index.html.

[6] C. Olsson et al. In-context learning and induction heads. *Transformer Circuits Thread*, 2022. https://transformer-circuits.pub/2022/in-context-learning-and-induction-heads/index.html.

[7] K. Li et al. Emergent world representations: Exploring a sequence model trained on a synthetic task. In *ICLR*, 2023. arXiv:2210.13382.

[8] A. M. Saxe, J. L. McClelland, and S. Ganguli. A mathematical theory of semantic development in deep neural networks. *PNAS*, 116(23):11537–11546, 2019.